HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Authored **hyperMILL** CAM programs for complex **5-axis components**, meticulously defining machining strategies to meet stringent aerospace and medical device industry tolerances.
- Developed clear **machine set-up instructions** and detailed machining programme notes for CNC operators, ensuring accurate and efficient production runs.
- Executed programme simulations and **verified completed programmes** against engineering drawings, identifying potential clashes or inefficiencies before production.
- Selected appropriate **cutting tools**, **fixturing solutions**, and **stock sizes**, integrating these choices into the overall machining strategy to optimize cycle times.
- Collaborated with **CNC operators** to gather feedback on programme performance and machine behaviour, iteratively modifying programmes to enhance efficiency and reduce waste by approximately **15%**.
- Reviewed and revised existing CNC programmes, implementing optimization techniques that improved machining accuracy and reduced tool wear by an estimated **10%** on legacy parts.
- Managed the transfer of newly developed programmes into the production environment, coordinating with manufacturing engineers to ensure a seamless transition and adherence to AS9100 and ISO 9001 standards.

SKILLS

CAD/CAM Software – hyperMILL, 2D modelling, 3D modelling, drafting.

CNC Machining – 5-axis programming, CNC machine set-up, fixturing, tooling selection, machining operations.

Engineering Drawing Interpretation – Reading and interpreting detailed engineering drawings.

Manufacturing Processes – Precision engineering, component manufacturing, AS9100, ISO 9001.

Programming & Automation – Python scripting, automation, data pipelines, ETL.

Databases & Cloud – SQL (PostgreSQL, MySQL, Oracle), AWS (EC2, S3, Lambda).

PROJECTS

Advanced 5-Axis Component Machining Strategy (HCL Technologies)

- Developed comprehensive machining strategies for intricate **5-axis components** using **hyperMILL**, focusing on complex surface contours and tight geometric tolerances.
- Created detailed toolpath simulations and verification reports, identifying and rectifying potential collision issues and optimizing cutting strategies to reduce cycle times by up to **20%**.
- Authored technical documentation for programme transfer, including tool lists, fixture requirements, and operational notes, supporting a smooth transition to the production floor.

Python Automation for Manufacturing Data (HCL Technologies)

- Engineered Python scripts to automate the processing of manufacturing data, extracting key metrics from production logs for analysis and reporting.
- Developed data pipelines to consolidate information related to tooling performance and machine uptime, facilitating a **15%** improvement in predictive maintenance scheduling.
- Integrated scripts with **PostgreSQL** databases to store and retrieve processed manufacturing data, enabling more efficient ad-hoc querying by operational teams.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)